I Equations et incertitudes selon [1]

"The main assumption of the shallow water model is that the horizontal length scale is much greater than the depth scale. Thus, one can get rid of the vertical dimension by averaging the mass and momentum conservation equations over the depth. In one dimension of space, the shallow water equations read" [1] - page VIII

"For such a problem, where shocks can form in the solution, finite volume methods have proved to be very effective." [1] - page VIII

"Generally, a robust and well-balanced scheme for shallow water equations should satisfy some mathematical properties such as preserve the nonnegativity of water height h, well-balanced, i.e. it preserves the discrete steady state, consistent with the shallow water equations and satisfy a discrete entropy inequality. A complete sta bility and convergence analysis of a finite volume scheme for a nonlinear hyperbolic system is out of reach. However, a number of properties of the scheme related to the stability and the convergence can be studied: preservation of invariant domains, en tropy inequalities, preservation of some particular steady states (well-balanced prop erty). Unfortunately, the literatures to study the staggered finite volume method for shallow water equations satisfy those properties are limited. Particularly, the study of staggered finite volume method in entropy satisfying is not available. Therefore through this thesis, we will try to investigate, design and analyze such properties of a staggered finite volume scheme for the shallow water equations with topography." - xiV

Voir Annexe A pour des solutions analytique qui permettront de tester le schéma 1D [1].

Inégalité avec une fonction entropie. -¿ Possible vérification ?

$$\partial_t h + \partial_x(hu) = 0$$
$$\partial_t(hu) + \partial_x(hu^2 + \frac{1}{2}gh^2) + gh\partial_x z = 0$$

Schéma explicite pour l'équation de conservation de la masse :

$$h_i^{n+1} = h_i^n - \frac{\Delta t}{\Delta x}(q_{i+\frac{1}{2}}^n - q_{i-\frac{1}{2}}^n)$$

avec $q_{i+\frac{1}{2}}^n = \hat{h}_{i+\frac{1}{2}}^n u_{i+\frac{1}{2}}^n$ et :

$$\hat{h}_{i+\frac{1}{2}}^n = \left\{ \begin{array}{ll} h_i^n & u_{i+\frac{1}{2}}^n \geq 0 \\ h_{i+1}^n & u_{i+\frac{1}{2}}^n < 0 \end{array} \right\}$$

Schéma explicite pour l'équation de la vitesse :

$$h_{i+\frac{1}{2}}^{n+1} u_{i+\frac{1}{2}}^{n+1} = h_{i+\frac{1}{2}}^n u_{i+\frac{1}{2}}^n - \frac{\Delta t}{\Delta x}(q_{i+1}^n \hat{u}_{i+1}^n - q_i^n \hat{u}_i^n \frac{1}{2} g((h_{i+1}^{n+1})^2 - (h_i^{n+1})^2) + g h_{i+\frac{1}{2}}^n (z_{i+1} - z_i))$$

avec $h_{i+\frac{1}{2}}^n = \frac{h_i^n + h_{i+1}^n}{2}$, $q_i^n = \frac{q_{i-\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n}{2}$ et

$$\hat{u}_i^n = \left\{ \begin{array}{ll} u_{i-\frac{1}{2}}^n & q_i^n \geq 0 \\ u_{i+\frac{1}{2}}^n & q_i^n < 0 \end{array} \right\}$$

Remarque : Le schéma est bien explicite : il faut d'abord calculer h puis avec sa valeur, on calcule u.

Valable sous la condition CFL suivante :

$$\nu = \frac{\Delta t}{\Delta x} \max_i (\frac{|q_{i+\frac{1}{2}}^n + q_{i-\frac{1}{2}}^n|}{2h_i^n} + \sqrt{g h_i^n}) < 1$$

Propriétés du schéma : Conservation de l'équilibre (Une solution à l'équilibre le reste)

Sources d'incertitudes : La profondeur z et la solution initiale. La masse volumique étant supposée constante dans un premier temps, son incertitude importe peu.

+ D'autres informations potentiellement utiles : Schéma en 2D, consistance, précision à l'ordre 2 (seulement 1D) et solutions analytique

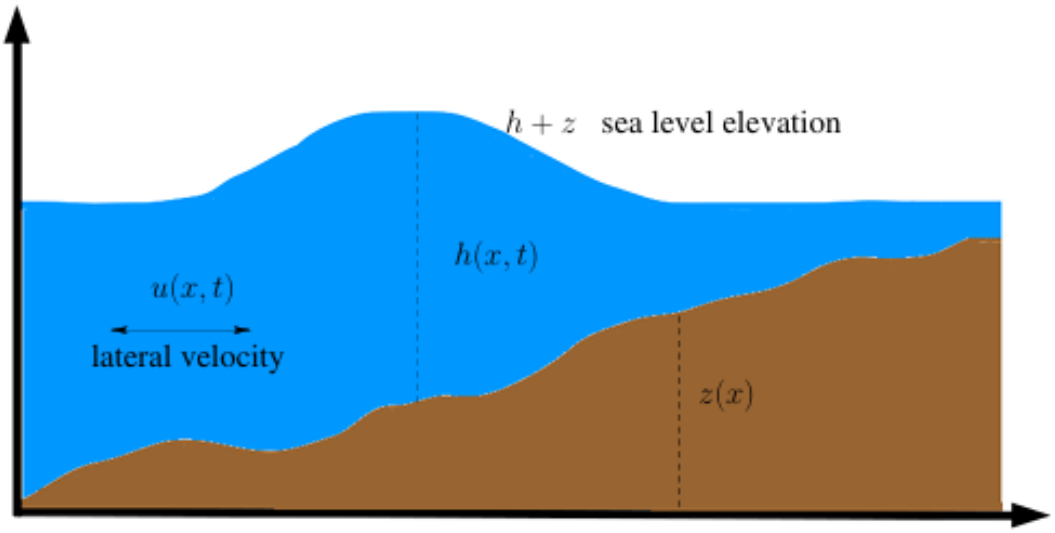


Figure 1: Shallow Water 1D (au sens de [1])

II Equations et incertitude selon [2]

Même équations.

$$U_i^{n+1} - U_i^n + \frac{\Delta t}{\Delta x}(F^n_{\frac{i+1}{2}} - F^n_{\frac{i-1}{2}}) = \Delta t S_i^n$$

Avec

$$S_i^n = \begin{pmatrix} 0 \\ \frac{g}{2}((max(0, h_{i,r} + z_{i,r} - max(z_{i,r}, z_{i+1,l})))^2 - (max(0, h_{i+1,l} + z_{i+1,l} - max(z_{i,r}, z_{i+1,l})))^2) \end{pmatrix}$$

$$F^n_{\frac{i+1}{2}} = \frac{f(U_i^n) + f(U_{i+1}^n)}{2} - c\frac{U_{i+1}^n - U_i^n}{2}$$

$$F^n_{\frac{i-1}{2}} = \frac{f(U_{i-1}^n) + f(U_i^n)}{2} - c\frac{U_i^n - U_{i-1}^n}{2}$$

$$U = (h, q)$$

$$f(U) = \begin{pmatrix} q \\ \frac{q^2}{h} + \frac{gh^2}{2} \end{pmatrix}$$

Bibliographie

[1] Harry Putu Gunawan. Numerical simulation of shallow water equations and related models. General Mathematics [math.GM]. Université Paris-Est; Institut teknologi Bandung, 2015. English.
`https://theses.hal.science/tel-01216642`.

[2] Yarodji, M.A.K., Badé, R. and Saley, B. (2020) Study of the 1D Saint-Venant Equations and Applica tion to the Simulation of a Flood Problem. Journal of Applied Mathematics and Phys ics, 8, 1193-1206.
`https://doi.org/10.4236/jamp.2020.87090`.